NARCIS: Authenticated Neural Coverless Image Signaling with Attack-Qualified Codebooks and Error Correction

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ABSTRACT

Selection-based coverless communication requires more than a stable image descriptor: the selected covers must jointly support the requested symbol sequence, survive channel distortions, and provide message-level security. This paper presents Neural Adaptive Robust Coverless Image Signaling (NARCIS), an end-to-end protocol that transmits unchanged natural images. A compact self-supervised encoder maps images to normalised descriptors. A quantile codebook provides balanced symbols, while an explicit multi-attack qualification stage retains covers whose labels remain stable. Payloads and session metadata are protected with AES-GCM, symbols are assigned through a keyed locality-preserving permutation, and Reed–Solomon coding converts residual classification errors into complete-message recovery. Evaluation uses five deterministic partitions of BOSSBase and five of Caltech-101, covering grayscale 512×512 photographs and variable-size RGB images. Across 1,575 clean, calibration, and holdout message-condition trials, NARCIS recovered every authenticated plaintext. Mean symbol accuracy was 98.74% on BOSSBase and 98.59% on Caltech-101; the worst individual accuracies were 81.61% and 79.89%. BOSSBase supported 3–4 bits per cover, whereas all Caltech-101 partitions supported 4 bits per cover. Selection bias was measured with three detector classes. On BOSSBase, all confidence intervals include chance performance. On Caltech-101, classical detectors identified a reproducible selection shift (global logistic AUC 0.531, 95% CI [0.504, 0.557]; residual ExtraTrees 0.533, [0.504, 0.562]); 20 repeated CNN fits on the most sensitive partition confirmed a partition-specific signal (mean AUC 0.533, 95% interval [0.517, 0.548]). On a matched BOSSBase seed-11 ablation, removing attack qualification reduced complete-message success from 100% to 67.6%; a calibration-locked projection search recovered 210/210 trials while increasing the minimum stable bucket from 92 to 161. Net plaintext rate increased from 0.184 to 1.213 bits per cover when the plaintext grew from 8 to 128 bytes at a 16-symbol operating point. The contribution is a measurable feasible operating point for authenticated coverless communication, defined by finite-index stability, protected-message demand, and exact end-to-end recovery.

1. Introduction

Coverless image steganography communicates information by selecting or generating natural visual content rather than by modifying an existing carrier [1, 2]. Selection-based methods associate image features with message symbols and retrieve a sequence of unchanged images from a shared index [3, 4]. Three interconnected gaps prevent these systems from providing end-to-end authenticated communication.

Gap 1: Finite-index message feasibility is not verified before transmission. Published methods report nominal hash length or average symbol accuracy. Neither metric guarantees that the symbol sequence required by a specific protected message can be realised from the available, qualified index. A transmission that cannot be realised fails silently, producing message-dependent coverage gaps invisible in per-symbol statistics [1, 2, 5].

Gap 2: No end-to-end selection-based protocol integrates authentication, replay control, and error correction. Selection-based schemes argue that cover integrity replaces steganographic hiding, but leave payload confidentiality, metadata authenticity, replay rejection, and channel error recovery as open problems [3, 4, 6, 7].

Gap 3: Cover stability filtering creates unquantified selection bias. Retaining only stable covers changes the distribution of transmitted images relative to the full index. Whether this selection leaks detectable information depends on the dataset, the filtering policy, and the detector, and this three-way dependence is rarely measured [8, 9].

This paper studies the following operational question:

> Given a finite shared image index, a fixed descriptor and codebook construction, a specified single-transformation channel, and a protected payload demand, what is the largest empirically feasible symbol alphabet for complete plaintext recovery using unchanged covers?

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The proposed protocol, Neural Adaptive Robust Coverless Image Signaling (NARCIS), answers this question by selecting its operating point from measured cover stability and message feasibility. The method learns a channel-invariant descriptor, partitions its principal direction with quantile boundaries, qualifies covers against a calibration attack set, and checks the multiplicity required by the protected message before transmission. Cryptographic and coding mechanisms are then integrated into the same end-to-end path. The answer is conditional on the evaluated index partition, descriptor, candidate codebooks, protected envelopes, and declared transformation sets; it is not a corpus-independent capacity statement.

NARCIS closes all three gaps with three corresponding contributions.

Contribution 1—Feasible operating point. A codebook is accepted only when every attack-qualified bucket satisfies the multiplicity required by the complete protected coded message. Across 1,575 end-to-end trials on BOSSBase and Caltech-101 under 21 channel conditions (including 8 holdout strengths), NARCIS recovered 100% of authenticated plaintexts (mean symbol accuracies 98.74% and 98.59%). Removing qualification on BOSSBase seed 11 reduced complete-message success from 210/210 to 142/210 (67.6%), with crops and large rotations failing completely.

Contribution 2—End-to-end cryptographic and coding integration. AES-GCM [10] authenticates payload and session metadata; associated data binds sender and receiver identities; keyed Gray permutation limits public symbol exposure; monotonic sequence checking rejects replay; Reed–Solomon [11] corrects residual label errors. Two formal propositions ground these properties in AES-GCM's established guarantee (Section 3). Net throughput at K=16 scales from 0.184 to 1.213 bits per cover for 8- to 128-byte payloads.

Contribution 3—Systematic detectability measurement. Three detector classes (global logistic regression, residual-feature ExtraTrees, selection CNN) are applied to five partitions of each dataset. BOSSBase AUC confidence intervals include 0.5. Caltech-101 shows a weak partition-specific CNN signal (mean AUC 0.533, 95% CI [0.517, 0.548] over 20 repeated fits on the most sensitive partition), reported transparently rather than suppressed.

The contribution concerns verified protected-message recovery at measured operating points under the stated datasets, attacks, detector classes, and threat model; it is not a claim of maximum nominal capacity.

2. Related Work

2.1. Hash- and retrieval-based coverless communication

Early coverless systems mapped robust image hashes to message fragments [1, 3]. Later retrieval-based schemes used local descriptors to improve resistance to common processing [4]. These methods preserve the transmitted image, but the database must contain sufficiently many stable representatives for every message symbol.

Abdulsattar [5] increased single-image nominal capacity by dividing a grayscale cover into overlapping blocks and mapping eigenvalue comparisons to ASCII values. WYSAWIS [12] instead groups 15-bit block hashes into 16 sequence categories and uses cloud-file rankings to represent segments. Both works illustrate that a large nominal representation space does not by itself guarantee that a given cover or index contains every required symbol with sufficient multiplicity.

2.2. Learned, joint, and stability-aware methods

Neural feature mapping and generation can provide semantic representations or larger message spaces [2, 13]. Selection and generation have been combined with StarGAN [14], while information-driven adversarial generation has been used to control the mapping from messages to synthetic content [15]. Recent semantic approaches derive symbols from neglected objects, human poses, or object detections [16–18]. Other systems use semantic-controlled text-to-image generation or latent diffusion to enlarge the generative cover space [19, 20]. These methods strengthen semantic capacity or cover generation, but they do not make finite-index feasibility, authenticated framing, and complete protected-message recovery interchangeable evaluation criteria.

Ren and Wu [21] combined selection and generation modules to address mapping completeness and reconstruction. Guo et al. [6] studied robustness and diversity against passive and active analysis. Guo and Ping [7] used pseudo-Zernike moments, stability-aware quantisation, and stability-regularised clustering. They reported average robustness of 99.54%, 98.64%, and 97.19% at 10, 14, and 15 bits on Holidays, VOC, and ImageNet, respectively.

NARCIS shares the stability-aware objective but differs in scope. Its primary unit of evaluation is the recovered protected message after classification, error correction, and authenticated decryption. It also selects the codebook size from finite-bucket feasibility. A direct numerical comparison with [7] is not attempted because the datasets, attack sets, and protocol overheads differ; the cited results serve as qualitative context only.

Deep image-hiding systems based on adversarial training, encoder–decoder networks, and invertible networks have established strong payload and reconstruction results [22–27]. These systems modify or generate carrier pixels and therefore address a different design space from selection-based coverless signalling. They remain relevant to evaluation practice because they analyse robustness, detectability, and decoding jointly rather than through nominal capacity alone.

2.3. Security and evaluation

No pixel modification removes embedding artefacts, but it does not by itself establish communication security. Selection may alter the distribution of transmitted content,

Table 1
Structural comparison of recent coverless image-steganography families. "Protocol protection" denotes integrated authenticated payload handling and error correction rather than image-level robustness alone.

Family	Representative works	Cover formation	Mapping or descriptor	Protocol protection
Block/hash mapping	Abdulsattar [5]; WYSAWIS [12]	Existing natural images	Block comparisons or category hashes	Not central to the cited mapping mechanism
Retrieval and local features	Li et al. [4]	Selected natural images	SURF-based retrieval	Not central to the cited retrieval mechanism
Adversarial generation	Qin et al. [13]; Chen et al. [14]; Zhang et al. [15]	Generated or selected/generated images	Latent or information-driven neural mapping	Varies by system; evaluated mainly at image/message mapping level
Semantic mapping	Zou et al. [16]; Tan et al. [17]; Meng et al. [18]	Existing natural images	Objects, poses, or detector outputs	Not the primary contribution of the cited methods
Text-to-image and diffusion	Li et al. [19]; Guo et al. [20]	Generated images	Semantic prompts or latent transformations	Not directly comparable to finite-index retrieval
Stability-aware selection	Guo et al. [6]; Guo–Ping [7]	Selected natural images	Stable handcrafted features and clustering	Image-level robustness and diversity
NARCIS	This work	Selected unchanged natural images	Invariant neural descriptor and qualified quantile codebook	AES-GCM, replay control, Reed–Solomon, and exact recovery

metadata can reveal protocol state, and an active adversary may alter or replay images. Steganalysis research has shown that weak statistical differences can be exploitable [8, 28–32]. Formal steganographic security also distinguishes distributional secrecy from cryptographic payload protection [9]. Accordingly, NARCIS separates cryptographic security mechanisms from empirical selection detectability. The latter is treated as a measured property of a specified detector, not as an unconditional security guarantee.

Table 1 separates these families by cover formation, mapping mechanism, and protocol-level protection.

3. System and Threat Model

The sender and receiver share an indexed corpus $\mathcal{I} = \{(x_i, y_i)\}_{i=1}^{N}$, a symbol-assignment key, and authenticated-encryption keys. Each x_i is a natural image and $y_i \in \{0, \dots, K-1\}$ is its codebook label. The sender transmits a sequence of unchanged images. The receiver applies the same descriptor and codebook to recover labels and then decodes the protected payload.

For each evaluated transmission, the channel applies at most one transformation from the specified set: JPEG compression, additive Gaussian noise, Gaussian blur, resizing, central cropping, or a small rotation. Attack compositions, repeated transformations, and adaptive multi-stage channels are not included in the present model. Calibration transformations and their parameters are available during index qualification; holdout strengths are excluded from qualification and used only for evaluation. The adversary knows the algorithm and may observe, alter, reorder, or replay transmissions, but does not know the shared keys and does not compromise the local index. Traffic analysis and semantic inference from image content remain outside the cryptographic claim.

The protocol targets:

1. *cover integrity*: no embedding or pixel modification by the sender;
2. *message correctness*: exact plaintext recovery or explicit failure;
3. *confidentiality and authenticity*: authenticated encryption of payload and session metadata;
4. *freshness*: rejection of repeated or out-of-order sequence numbers;
5. *channel robustness*: correction of residual label errors within the configured Reed–Solomon budget.

3.1. Formal security properties

The following propositions state the cryptographic guarantees provided by NARCIS under standard assumptions. They follow directly from the authenticated encryption security of AES-GCM [10] and require no new cryptographic proof.

Proposition 1 (Payload Confidentiality)**.** *For plaintext m, nonce derived from sequence number n, and key k_p unknown to the adversary, the ciphertext* $c = \mathrm{AEAD.Enc}_{k_p}(m; \text{nonce}, \text{AD})$ *is computationally indistinguishable from a uniformly random string of the same length, with advantage bounded by the IND-CCA2 security of AES-GCM [10]. An adversary observing transmitted images and sealed metadata gains no information about m.*

Proposition 2 (Authenticity and Replay Rejection)**.** *A message is accepted by the receiver only when (i) the AES-GCM tag on the sealed metadata verifies under k_m with* *sender–receiver identities as associated data, and (ii) the sequence number n satisfies $n > n_{\max}$, where $n_{\max}$ is the highest previously accepted value for that sender. Under AES-GCM's authentication guarantee [10], any altered or replayed message fails condition (i) or condition (ii).*

Traffic analysis and semantic inference from image content remain outside the cryptographic claim. Selection detectability is treated as a measured empirical property (Section 6).

4. NARCIS Method

4.1. Channel-invariant neural descriptor

Channel transformations are applied at the declared channel resolution, and the resulting images are resized to 128×128 for descriptor inference. The encoder contains four convolutional blocks with 16, 32, 64, and 96 channels. Each block uses two 3×3 convolutions, batch normalisation, and SiLU activations; the last three blocks downsample by a factor of two. Global average pooling and a two-layer $96 \rightarrow 128 \rightarrow 64$ projection head produce a 64-dimensional unit-normalised descriptor $z = f_\theta(x)$. The grayscale and RGB instances have 231,312 and 231,600 trainable parameters, respectively; only the first convolution differs. Figure 1 gives the implemented dimensions.

Training uses two independently distorted views $x^{(1)}$ and $x^{(2)}$ of the same image. Each view is generated by randomly selecting JPEG compression, Gaussian noise, blur, resize, crop, rotation, or the identity transform. The loss is

$$\mathcal{L} = \lambda_{\text{inv}}\mathcal{L}_{\text{inv}} + \lambda_{\text{var}}\mathcal{L}_{\text{var}} + \lambda_{\text{cov}}\mathcal{L}_{\text{cov}}, \tag{1}$$

where $\mathcal{L}_{\text{inv}}$ is the mean squared distance between paired descriptors, $\mathcal{L}_{\text{var}}$ prevents dimensional collapse, and $\mathcal{L}_{\text{cov}}$ penalises off-diagonal covariance. This variance–invariance–covariance structure follows the non-contrastive principle used by VICReg [33]; unlike contrastive objectives such as SimCLR [34], it does not require negative pairs. The weights are 25, 25, and 1, and the target standard deviation is $64^{-1/2}$, consistent with unit-normalised descriptors. A targeted sensitivity study in Section 5.5 compares these weights with two alternatives and dimensions 32 and 128. No semantic labels are used.

For a batch of B paired d-dimensional descriptors Z and Z', the three terms are

$$\mathcal{L}_{\text{inv}} = \frac{1}{Bd}\|Z - Z'\|_F^2, \tag{2}$$

$$\mathcal{L}_{\text{var}} = \frac{1}{d}\sum_{j=1}^{d}\left[\max(0, \gamma - \sigma_j(Z)) + \max(0, \gamma - \sigma_j(Z'))\right], \tag{3}$$

$$\mathcal{L}_{\text{cov}} = \frac{1}{d(d-1)}\sum_{i\neq j}\left(C_{ij}(Z)^2 + C_{ij}(Z')^2\right), \tag{4}$$

where $\gamma = d^{-1/2}$, σ_j is the batch standard deviation with 10^{-4} variance regularisation, and $C(Z) = (Z - \bar{Z})^\top(Z - \bar{Z})/(B-1)$. These definitions match the released implementation.

4.2. Balanced quantile codebook

Let $Z \in \mathbb{R}^{N\times 64}$ contain clean descriptors and let v_1 be the first right singular vector of the centred matrix. Each descriptor is projected as

$$s_i = (z_i - \bar{z})^\top v_1. \tag{5}$$

For a candidate alphabet of size K, boundaries are placed at the empirical quantiles j/K, $j = 1, \ldots, K-1$. The resulting label is

$$q_K(z_i) = \sum_{j=1}^{K-1}\mathbf{1}[s_i > b_j]. \tag{6}$$

Quantile partitioning balances clean occupancy before channel qualification and avoids empty clusters caused solely by an unconstrained fit. The complete clean-image assignment is defined by Eq. (6). The first singular direction is used because it is deterministic and captures the largest descriptor variance with one scalar. In the reduced sensitivity subset it explained 72.90% of the variance; the first two directions jointly explained 96.83%. Alternative directions occasionally retained more stable covers, so v_1 is treated as a reproducible design choice rather than a stability optimum. A targeted full-index study therefore evaluates a calibration-locked attack-aware alternative. Its candidate set contains the first 16 principal directions and 32 deterministic random unit directions. For each candidate, quantile boundaries and stable buckets are computed using clean images and calibration attacks only. Selection lexicographically maximises the minimum stable bucket and retained-cover count, then minimises occupancy variation. The chosen direction is fixed before holdout evaluation.

4.3. Attack-qualified cover index

A cover is retained only when its label is unchanged under every calibration condition:

$$\mathcal{S}_K = \left\{ i : q_K(f_\theta(T(x_i))) = q_K(f_\theta(x_i)),\ \forall T \in \mathcal{T}_{\text{cal}} \right\}. \tag{7}$$

The calibration set contains 12 transformations: JPEG qualities 80 and 50; Gaussian noise with standard deviations 5 and 12; Gaussian blur radii 0.8 and 1.5; resize factors 0.75 and 0.50; crop fractions 0.05 and 0.10; and rotations of 3° and 7°.

Stability filtering can introduce visible selection bias. Seven image statistics are therefore used to estimate the probability that a cover is stable: mean, standard deviation, horizontal and vertical gradient magnitude, entropy, and the 0.1 and 0.9 quantiles. Within each symbol bucket, stable covers are ordered by clipped inverse-propensity weighted random priorities. This ordering does not change labels or pixels; it reduces deterministic preference for visually distinctive stable covers.

4.4. Protected message representation

For plaintext m and sequence number n, AES-GCM [10] produces

$$c = \text{AEAD.Enc}_{k_p}(m; \text{nonce}, \texttt{NARCIS-PAYLOAD-1}\|n). \tag{8}$$

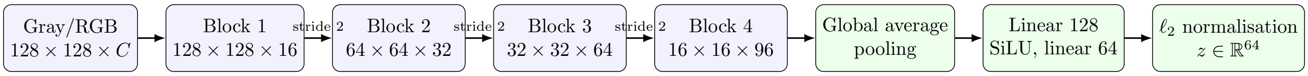

Figure 1: Descriptor encoder used in both principal evaluations, with $C = 1$ for BOSSBase and $C = 3$ for Caltech-101. Each convolutional block contains two 3×3 convolution–batch normalisation–SiLU stages. Channel transformations precede model resizing.

The ciphertext envelope is framed with a protocol marker, length, and CRC-32, then encoded with Reed–Solomon parity [11]. Session metadata contains the sequence number, padding, codebook size, and cover count. It is independently sealed with AES-GCM using sender and receiver identities as associated data. The receiver accepts only a sequence number greater than the highest previously accepted value for that sender.

For $K = 2^b$, the coded bit stream is divided into b-bit symbols. A keyed permutation combines a Gray ordering, a key-derived cyclic shift, and a key-derived orientation. Adjacent symbol values consequently remain adjacent in the unshifted codebook order, while the externally observed symbol-to-label assignment depends on the key.

More precisely, let $h = \mathrm{HMAC\text{-}SHA256}(k_s, \texttt{cluster-order})$ [35], $a = \mathrm{int}(h_{0:8}) \bmod K$, and $r = -1$ if the least significant bit of h_8 is one, otherwise $r = 1$. For position $p \in \{0, \dots, K-1\}$, define the binary-reflected Gray symbol $g(p) = p \oplus (p \gg 1)$ and

$$\pi_k(g(p)) = (a + rp) \bmod K. \tag{9}$$

The sender maps each coded symbol through π_k; the receiver constructs the exact inverse table. Equation (9) fully specifies the mapping and separates its role from payload encryption.

4.5. Feasible operating point

Let $d_k(c)$ denote the number of occurrences of label k required to transmit protected envelope c, and let a_k be the number of unused qualified covers in bucket k. A candidate codebook is feasible when

$$d_k(c) \le a_k, \qquad \forall k,\ \forall c \in \mathcal{C}_{\mathrm{eval}}. \tag{10}$$

Candidate sizes are evaluated from the largest to the smallest. The selected operating point for a partition is the largest K satisfying Eq. (10) for every protected envelope generated before channel evaluation for that partition. This single partition-level value is then fixed for all messages and all 21 channel conditions. No message-specific or attack-specific reselection is performed. The experiment therefore establishes feasibility for the evaluated envelope set $\mathcal{C}_{\mathrm{eval}}$; it does not guarantee every possible message up to a nominal length. This criterion distinguishes nominal symbol capacity $\log_2 K$ from the net payload rate after encryption, framing, and error correction.

4.6. Computational complexity

Let N be the index size, $A = |\mathcal{T}_{\mathrm{cal}}|$, d the descriptor dimension, K the alphabet size, L the number of transmitted covers, and F_θ the cost of one encoder inference.

Algorithm 1: NARCIS protected coverless transmission

Input: Plaintext m, sequence n, qualified index $\mathcal{I}$, keys
Output: Recovered plaintext or failure

Encrypt m with sequence-bound AES-GCM;
Frame the ciphertext and apply Reed–Solomon coding;
Use the partition-level codebook K selected before channel evaluation;
Map coded symbols to labels with the keyed permutation;
Select one unused qualified cover for each required label;
Transmit the unchanged cover sequence and sealed metadata;
Verify metadata and reject replayed sequence numbers;
Classify received covers and invert the keyed mapping;
Apply Reed–Solomon decoding and verify CRC;
Authenticate and decrypt the payload;
return plaintext if every verification succeeds; otherwise failure;

Clean embedding costs $O(NF_\theta)$. Multi-attack qualification is the dominant offline term, $O(ANF_\theta + ANKd)$ when label assignment is written explicitly. The scalar quantile implementation reduces label assignment after projection to $O(AN \log K)$ through binary search over the ordered boundaries. PCA fitting costs $O(Nd^2)$ for $N \ge d$, and sorting the projected values costs $O(N \log N)$. Online cover retrieval and keyed mapping are $O(L)$ with pre-built buckets, while decoding adds the Reed–Solomon block cost. Storage is $O(Nd + N)$ for descriptors and indexed identifiers. Offline qualification is therefore intentionally amortised across subsequent transmissions.

Figure 2 separates offline index construction from online protected transmission and recovery.

Algorithm 1 summarises transmission and recovery.

5. Experimental Protocol

5.1. Dataset and partitions

The principal evaluation uses BOSSBase 1.01 [36], containing 10,000 grayscale photographs at 512×512 pixels. The dataset is kept at native resolution for JPEG, noise, blur, resize, crop, and rotation; the resulting image is resized to

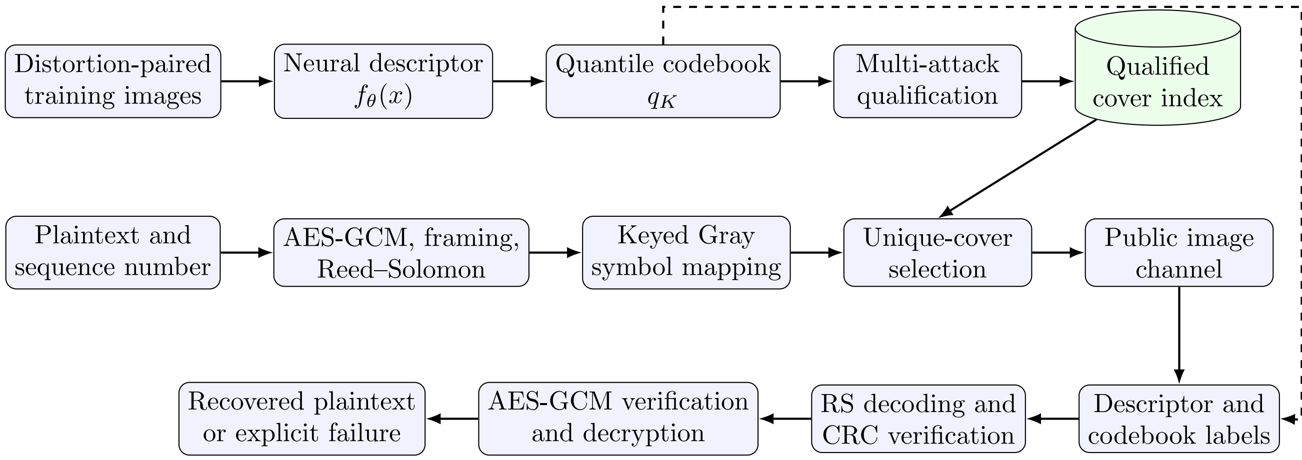

Figure 2: NARCIS construction and transmission pipeline. Training and index qualification are performed before communication. The transmitted objects are unchanged images selected from the qualified index.

128×128 only at the encoder boundary. For each seed in $\{11, 29, 47, 71, 101\}$, a deterministic permutation provides 2,000 training images and a disjoint 8,000-image cover index.

Cross-dataset validation uses Caltech-101 [37]: 9,144 natural RGB images from 101 object categories with variable source resolutions. The channel input is centre-fitted to 256×256 in RGB, and the transformed image is resized to 128×128 only at the encoder boundary. Each of the same five seeds assigns 1,500 images to descriptor training and 7,000 disjoint images to the shared index. Semantic labels are neither used in training nor in codebook construction.

The encoder is trained separately on each partition for five epochs with AdamW, batch size 16, learning rate 10^{-3}, and weight decay 10^{-5}. The BOSSBase experiment evaluates ten independently generated eight-byte plaintexts per seed; Caltech-101 evaluates five. Reed–Solomon coding uses 128 parity bytes. Symbol repetition is set to one; redundancy is instead provided by byte-level Reed–Solomon parity, which avoids multiplying the required 348–464-cover sequences. A separate deterministic scaling analysis encodes 8-, 32-, and 128-byte plaintexts with the implemented AES-GCM envelope, 10-byte frame, and 128 parity symbols per Reed–Solomon block. The two principal evaluations comprise 75 protected messages and 1,575 message-condition trials. For each partition, all protected envelopes are generated before channel evaluation and jointly determine one feasible operating point. That value is held constant across the clean channel, the 12 calibration conditions, and the eight holdout strengths.

CIFAR-100 [38] is retained only for the controlled qualification ablation and the matched local-descriptor comparison reported in Section 6.8. That experiment uses three seeds, 2,000 training images, an 8,000-image index, and three messages per seed.

5.2. Channel conditions and metrics

The 12 calibration transformations are those in Eq. (7). Eight holdout strengths are used only at evaluation: Gaussian noise with standard deviations 9 and 15; blur radii 1.2 and 1.8; crop fractions 0.08 and 0.12; and rotations of 5° and 9°. Together with the clean channel, this gives 21 conditions. Each condition applies one transformation independently to the clean cover; the protocol is not evaluated under composed or sequential distortions.

Two metrics are reported:

1. *symbol accuracy*, the fraction of transmitted covers assigned to their intended labels after channel processing;
2. *complete-message success*, which is one only when error correction, CRC verification, authenticated decryption, and plaintext equality all succeed.

Confidence intervals are two-sided 95% Student-*t* intervals over the five seed-level values. They quantify variation across deterministic partitions; they do not imply coverage over datasets or untested channels.

5.3. Baselines and ablations

The local baselines use a normalised 16×16 low-frequency DCT descriptor and a normalised 64-bin grayscale histogram. They are subjected to the same quantile partition, calibration attacks, candidate codebook sizes, and protected-message feasibility rule as NARCIS. This is a controlled descriptor comparison, not a reproduction of a published complete protocol.

The principal-corpus ablation removes attack-based cover qualification on BOSSBase seed 11 while retaining the trained descriptor, 16-symbol PC1 codebook, ten protected messages, 128 parity bytes, and all 21 channel conditions. A smaller three-seed CIFAR-100 ablation provides an independent controlled replication. A second check confirms the configured repetition factor of one.

5.4. Selection detectability

A deliberately interpretable detector uses seven global image statistics: mean, standard deviation, two gradient magnitudes, entropy, and two intensity quantiles. Logistic regression distinguishes selected covers from an equal number of non-selected index images using a stratified 65/35 train/test split. A second detector extracts 52 high-pass residual co-occurrence summaries inspired by rich-model steganalysis [8] and applies an ExtraTrees ensemble. A third detector is a lightweight selection CNN with three convolutional stages (16, 32, and 64 channels), batch normalisation, spatial averaging, and a binary output. It is trained for eight epochs on 64×64 views using the same split. The area under the ROC curve (AUC) measures detectability. These tests probe qualification-induced selection bias at increasing model flexibility. The CNN is not presented as a reproduction of XuNet or SRNet [29, 32], whose original task is detecting pixel embedding rather than unchanged-image selection. Because the initial Caltech-101 CNN result varied across partitions, seed 47 is analysed with 20 independent balanced subsamples, train/test splits, and CNN initialisations. Standardised differences in the seven image statistics and category-frequency shifts are also reported to identify plausible sources of selection leakage.

5.5. Sensitivity and computational measurements

A targeted reduced experiment uses seed 11, 500 training images, a 1,500-image index, three epochs, and 96×96 model inputs. It compares descriptor dimensions 32, 64, and 128; loss weights $(25, 25, 1)$, $(1, 1, 1)$, and $(25, 10, 1)$; and projections on the first three principal directions and a fixed random direction. This experiment diagnoses design choices and is not used to replace the five-seed operating point.

A second projection study uses the complete 8,000-image BOSSBase seed-11 index and its released checkpoint. It evaluates the 48 predeclared candidate directions described in Section 4.2 under all 12 calibration attacks, locks the selected direction, and then executes ten messages under the complete 21-condition suite. Explained-variance–stability correlation is computed only across the 16 principal directions.

Wall-clock measurements include native-image descriptor extraction, calibration, and complete transmission and recovery. Parameter counts are reported directly from the implemented models.

6. Results

6.1. Codebook feasibility

Table 2 reports the selected BOSSBase configurations. Four partitions supported 16 symbols, corresponding to 4 bits per cover. Seed 47 supported eight symbols, or 3 bits per cover. Between 30.38% and 48.73% of the index remained stable under all 12 calibration attacks.

Each protected eight-byte plaintext required 348 covers at 4 bits per cover and 464 covers at 3 bits per cover. The measured net plaintext rates were therefore 0.184 and 0.138 bits per transmitted cover. These values include encryption, framing, and error correction and must not be confused with the nominal alphabet.

Table 2
Selected BOSSBase operating points on 8,000-image indices.

Seed	K	Bits/cover	Stable covers	Min. bucket
11	16	4	2,823 (35.29%)	92
29	16	4	2,430 (30.38%)	61
47	8	3	3,898 (48.73%)	177
71	16	4	2,687 (33.59%)	80
101	16	4	2,846 (35.58%)	99

Table 3
Implemented payload scaling with 128 RS parity symbols per block.

Plaintext	RS blocks	Coded bytes	Covers ($K = 16$)	Net b/cover
8 bytes	1	174	348	0.184
32 bytes	1	198	396	0.646
128 bytes	2	422	844	1.213

6.2. Payload-overhead scaling

Table 3 and Figure 3 report the implemented coding path at three plaintext sizes. AES-GCM adds 28 bytes and the protocol frame adds 10. The 128-byte payload exceeds the 127-byte data capacity of one RS(255,127) block after encryption and framing; the library therefore emits two blocks and 256 effective parity bytes. Despite this block transition, the fixed overhead is increasingly amortised: at $K = 16$, net rate rises from 0.184 to 0.646 and 1.213 bits per cover.

6.3. End-to-end robustness

All 1,050 BOSSBase trials succeeded: five seeds, ten messages, and 21 conditions. Mean symbol accuracy was 98.74%. Figure 4 shows per-condition means. The two most difficult conditions were holdout crop 0.12 (84.37%) and holdout rotation 9° (91.75%). The worst individual symbol accuracy was 81.61%. Reed–Solomon corrected at most 61 byte positions, and authenticated decryption recovered every plaintext.

The clean channel, both JPEG settings, both resize settings, calibrated crops, both blur calibration settings, and rotations up to 3° produced 100% mean symbol accuracy. Message success is not inferred from these averages; it is verified after Reed–Solomon decoding, CRC checking, and authenticated decryption.

6.4. Cross-dataset validation

All five Caltech-101 partitions selected $K = 16$, corresponding to 4 bits per cover. Table 4 reports the retained-cover statistics. Stable fractions ranged from 22.00% to 34.74%, and every minimum bucket supported the five protected messages evaluated for its partition.

All 525 Caltech-101 message-condition trials recovered the authenticated plaintext. Mean symbol accuracy was

Table 4
Selected Caltech-101 operating points on 7,000-image RGB indices.

Seed	Bits/cover	Stable covers	Min. bucket
11	4	1,678 (23.97%)	38
29	4	1,540 (22.00%)	31
47	4	2,432 (34.74%)	76
71	4	2,289 (32.70%)	45
101	4	1,895 (27.07%)	35

Table 5
Selection-detection AUC (mean and 95% Student-t CI over five partitions). Intervals excluding 0.5 indicate a measurable shift.

Dataset	Global logistic	Residual ExtraTrees	Selection CNN
BOSSBase	0.518 [0.489, 0.547]	0.514 [0.486, 0.543]	0.515 [0.463, 0.566]
Caltech-101	0.531 [0.504, 0.557]	0.533 [0.504, 0.562]	0.530 [0.481, 0.579]

98.59%, the worst individual accuracy was 79.89% under the unseen 12% crop, and Reed–Solomon corrected at most 63 byte positions. Figure 5 compares the two datasets condition by condition. The same holdout crop and rotation remain the hardest conditions, while dataset-specific differences appear for strong blur.

6.5. Training convergence

Figure 6 reports the complete five-epoch histories. Loss decreased substantially beyond epoch two for every seed. The final values ranged from 0.066 to 0.157. Individual curves are not monotonic because each epoch samples new distortion pairs. The experiment supports the use of five epochs over the earlier two-epoch setting, without asserting asymptotic convergence.

6.6. Selection detectability

Table 5 reports the mean AUC and 95% Student-t confidence intervals over five partitions for each detector and dataset.

On BOSSBase, all three confidence intervals include 0.5. On Caltech-101, the two classical detectors identify a small dataset-specific selection shift (intervals exclude 0.5), while the CNN interval includes 0.5 over five partitions. Figure 7 summarises these results.

The BOSSBase intervals and both CNN intervals include 0.5. The two classical Caltech-101 detectors show a small but measurable dataset-specific selection shift, while the more flexible CNN result is less precise and does not exclude chance. These outcomes motivate reporting selection detectability by dataset and detector rather than inferring a general undetectability property.

The targeted seed-47 analysis resolves part of that uncertainty. Twenty independent CNN subsamples, splits, and initialisations yielded mean AUC 0.533, 95% interval [0.517, 0.548], and range 0.445–0.580 (Figure 8). Thus, an adversary trained on a matched Caltech-101 index can exploit a weak selection signal on this partition. No individual global statistic had a standardised mean difference above 0.088. Category proportions shifted more strongly: motorbikes and `faces_easy` were under-represented by 0.022 and 0.020, whereas `background_google` and anchor were over-represented by 0.017 and 0.015. These observations indicate a mixture of weak visual and semantic composition effects rather than one dominant low-level feature.

Table 6
Targeted sensitivity study on the reduced BOSSBase subset.

Configuration	Stable fraction	Min. bucket
64D, 25/25/1	0.335	9
32D, 25/25/1	0.375	19
128D, 25/25/1	0.412	20
64D, 1/1/1	0.284	5
64D, 25/10/1	0.357	19

6.7. Sensitivity and complexity

Table 6 summarises the reduced sensitivity experiment. Equal loss weights retained fewer stable covers and produced the smallest minimum bucket. Dimension 128 improved retained stability in this subset, but a confirmatory two-seed full-protocol run recovered 419/420 trials, compared with 420/420 for the corresponding 64-dimensional runs. The 64-dimensional configuration is therefore retained because the larger representation did not improve tested capacity or reliability. The one-trial difference is not interpreted as statistically significant; it only shows that the larger descriptor provided no observed reliability gain in this limited confirmatory run.

PC1 explained 72.90% of descriptor variance and retained 33.47% of covers. PC2 retained 34.93%, PC3 31.40%, and a fixed random direction 41.27%. This result rules out a claim that PC1 is stability-optimal.

The full-index attack-aware study selected fixed random direction 15 using calibration data only. Table 7 compares the key candidate directions. PC1 and PC2 retain the same stable images (2,823) but PC2 achieves a higher minimum bucket (115 vs. 92). The 32 random directions achieve substantially higher stable fractions (median 45.4%) and the best (random_15) reaches minimum bucket 161. This counter-intuitive pattern arises because PC1 encodes the descriptor dimensions most perturbed by channel attacks; lower-variance or random directions encode less channel-sensitive content and retain more stable covers. Figure 9 shows all 48 candidates. Across the 16 principal directions, explained variance and stable fraction were positively correlated ($\rho = 0.640$, $p = 0.0076$). High variance therefore did not imply greater instability in this experiment; PC1 was suboptimal because variance maximisation and minimum qualified-bucket occupancy are different objectives. After the direction was locked, the attack-aware variant recovered 210/210 messages across all channel conditions, with 99.24% mean and 87.64% worst symbol accuracy.

Native-image descriptor extraction averaged 87.83 seconds for 8,000 covers, or 99.85 images/s. Calibration averaged 1,246.10 seconds per partition. Complete evaluation

Table 7
Projection candidates on the full BOSSBase seed-11 index (K=16, 8,000 images). PC1 and PC2 retain the same stable images but differ in bucket balance. The calibration-locked random direction achieves the highest minimum bucket.

Direction	Expl. var.	Stable images	Min. bucket
PC1	94.07%	2,823 (35.3%)	92
PC2	5.19%	2,823 (35.3%)	115
Random (median of 32)	—	3,630 (45.4%)	101
Random (best of 32)	—	3,867 (48.3%)	161

Table 8
BOSSBase seed-11 matched ablation: complete-message success over 10 messages per condition. "Qualif." = PC1-qualified (NARCIS standard); "No qualif." = PC1 without qualification. (H) = holdout strength.

Condition	Qualif.	No qualif.	Sym. acc. (no qual.)
Clean	10/10	10/10	1.000
JPEG 80, 50	10/10	10/10	>0.993
Gaussian σ5, 12	10/10	10/10	0.926–0.930
Gaussian σ9 (H), 15 (H)	10/10	10/10	0.925–0.926
Blur 0.8, 1.5	10/10	10/10	0.872–0.950
Blur 1.2 (H), 1.8 (H)	10/10	10/10	0.847–0.904
Resize 0.75, 0.50	10/10	10/10	0.936–0.959
Crop 5%, 10%	10/10	**0/10**	0.430–0.627
Crop 8% (H), 12% (H)	10/10	**0/10**	0.381–0.524
Rotate 3°	10/10	10/10	0.887
Rotate 7°	10/10	**0/10**	0.719
Rotate 5° (H)	10/10	**2/10**	0.788
Rotate 9° (H)	10/10	**0/10**	0.666
Total	**210/210**	**142/210 (67.6%)**	—

averaged 754.63 seconds for 210 message-condition trials, equivalent to 3.59 seconds per trial. Dimensions 32, 64, and 128 contain 218,928, 231,312, and 268,368 parameters, respectively; similar inference times confirm that the convolutional backbone dominates cost.

6.8. Secondary ablation and descriptor comparison

On BOSSBase seed 11, the matched PC1 protocol without qualification recovered 142/210 messages (67.62%), compared with 210/210 for the qualified PC1 protocol. Table 8 shows the per-condition breakdown. Success was zero under all crop conditions and rotations $\geq$ 7°; only 2/10 messages survived the 5° rotation. The remaining 14 conditions recovered 140/140 messages, confirming that invariance training alone tolerates standard distortions; qualification is necessary only for distortions with sharp spatial structure. The controlled CIFAR-100 experiment independently recovered 189/189 qualified trials and fell to 43.65% without qualification. Agreement across the principal corpus and the smaller controlled experiment identifies qualification as a necessary robustness component rather than treating invariant training alone as sufficient.

Under the same quantile partition, attack set, candidate sizes, and protected-message demand, neither the 16×16 DCT descriptor nor the 64-bin histogram formed a feasible codebook in any of three partitions. This is a controlled descriptor comparison, not a claim against all handcrafted coverless systems.

6.9. Position relative to published methods

Table 9 separates directly measured properties from published context. Abdulsattar and WYSAWIS provide high nominal intra-image or category-based representations [5, 12], whereas NARCIS evaluates a protected multi-cover sequence. Guo and Ping [7] report higher nominal capacities and high average robustness on larger datasets. Their reported capacity is therefore stronger than the 3–4 bit operating points measured here.

NARCIS's advantage is therefore not global capacity. Its supported positioning is protocol completeness: unchanged-cover signalling, explicit finite-index feasibility, attack-qualified selection, authenticated communication, replay control, and complete-message recovery under calibrated and holdout distortions.

7. Discussion

7.1. What the experiments establish

The experiments support five conclusions. First, stable descriptor averages are insufficient for communication: the multiplicity of every required symbol must be checked. Second, cover qualification is the dominant robustness component in the tested pipeline. Third, forward-error correction allows message recovery at symbol accuracies substantially below 100%. Fourth, cryptographic verification changes the evaluation criterion: a trial is successful only when the exact plaintext is authenticated and recovered. Fifth, short-message overhead is substantial but is measurably amortised as payload length increases, subject to Reed–Solomon block boundaries.

The five-seed results also quantify finite-index and cross-corpus variability. The same rules yield a feasible 16-symbol alphabet on four BOSSBase partitions and an eight-symbol alphabet on one, while all five Caltech-101 partitions support 16 symbols. Reporting the lower BOSSBase operating point avoids selecting only favourable partitions.

7.2. Security interpretation

AES-GCM, associated data, and monotonic sequence checking provide conventional confidentiality, authenticity, binding, and replay rejection under correct key and nonce management. These properties do not conceal communication volume or image semantics. Distributional detectability is therefore evaluated separately from cryptographic payload security. The AUC experiment addresses a different question: whether the current selection policy creates detectable shifts in seven global statistics, residual summaries, or a learned CNN representation. The classical-detector shift on Caltech-101 and the repeated seed-47 CNN analysis establish a weak but measurable selection leak on that partition. An adversary with access to a similarly distributed index and labelled selection traces can therefore exceed chance detection. These

Table 9
Positioning of NARCIS. Values from different publications are not treated as a common-protocol ranking.

Property	Abdulsattar [5]	WYSAWIS [12]	Guo–Ping [7]	NARCIS
Representation	Eigenvalue comparisons in overlapping blocks	15-bit block hashes grouped into 16 categories	PZM features with stability-aware quantisation and clustering	Learned invariant descriptors with quantile codebook
Primary capacity statement	High nominal single-cover location capacity	High nominal block/category capacity	10, 14, and 15 bits on Holidays, VOC, and ImageNet	3–4 bits on BOSSBase; 4 bits on Caltech-101
Explicit multi-attack cover qualification	Not evaluated here	Not evaluated here	Yes	Yes
Message-level error correction	Not part of the compared mechanism	Not part of the compared mechanism	Not stated in the reported context	Reed–Solomon, measured after complete decoding
Authenticated payload and metadata	Not part of the compared mechanism	Not part of the compared mechanism	Not stated in the reported context	AES-GCM with sequence-bound data and replay guard
Holdout attack strengths	Not evaluated here	Not evaluated here	Not established from the cited summary	Eight holdout strengths
Measured complete protected-message recovery	Not evaluated here	Not evaluated here	Not directly comparable	1,575/1,575 trials

measurements must remain tied to a dataset, partition, selection policy, and detector; they do not support a general undetectability claim.

7.3. Limitations

The evaluation has five main limitations.

1. Caltech-101 adds variable-resolution RGB content, but its channel input is centre-fitted to 256×256 for batching. Evaluation on larger native-resolution colour corpora would test a broader operating regime.
2. The local baselines isolate descriptor feasibility; they are not faithful reimplementations of complete published systems.
3. The learned selection detector is a lightweight CNN designed for the selected-versus-non-selected task, not an SRNet reproduction. Traffic-level sequence analysis is also outside this study.
4. The attack-aware projection result is a full-index single-partition validation. Applying the same calibration-locked rule across all partitions would be required before replacing PC1 in the principal five-seed results.
5. Caltech-101 seed 47 exhibits measurable selection leakage. The present inverse-propensity ordering reduces deterministic preference but does not equalise category composition or provide traffic-level indistinguishability.

The measured net plaintext rates, 0.184 and 0.138 bits per transmitted cover, are lower than the nominal symbol capacities because the experiments use short payloads and 128 parity symbols per RS block. The measured scaling curve reaches 1.213 bits per cover for a 128-byte plaintext at $K = 16$, while also showing the second-block overhead introduced beyond 127 framed bytes.

8. Conclusion

NARCIS integrates learned image descriptors, balanced codebooks, attack-based cover qualification, keyed assignment, authenticated encryption, replay control, and Reed–Solomon correction into one selection-based coverless protocol. It recovered 1,050 of 1,050 protected message-condition trials on BOSSBase and 525 of 525 on RGB Caltech-101, including eight holdout attack strengths. BOSSBase supported 3–4 bits per cover; every Caltech-101 partition supported 4 bits per cover. Mean symbol accuracies were 98.74% and 98.59%. Removing qualification in the controlled ablation reduced complete-message success to 67.62% on BOSSBase seed 11 and 43.65% in the independent CIFAR-100 experiment.

These results support a precise position: NARCIS provides measured robust operating points and complete protected-message recovery with unchanged natural covers. Operationally, the results show that finite-index feasibility and byte-level correction can convert imperfect label stability into exact, authenticated recovery without modifying transmitted images. The evidence does not support a global capacity or universal steganalysis claim.

Three extensions follow directly from the observed limits. First, the calibration-locked attack-aware projection rule should be evaluated across all partitions before replacing PC1 in the principal operating points. Second, executable recent-protocol baselines should be evaluated under the same payload protection, attacks, and finite-index demand.

Third, larger native-resolution colour corpora and traffic-level sequence detectors should extend the present image-level analysis; selection should additionally be balanced against semantic composition. These directions target capacity, comparative attribution, and detectability separately while preserving the feasible-operating-point criterion as the common evaluation unit.

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CRediT authorship contribution statement

Stéphane Gaël R. Ekodeck: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Vivien Beyala Kamgang:** Conceptualization, Methodology, Writing – review and editing. **Serge Alain Ebélé:** Conceptualization, Methodology, Validation, Writing – review and editing. **Julius Marcellin Nkenlifack:** Supervision, Project administration, Writing – review and editing.

Declarations

Ethics approval. This study does not involve human participants, animals, personal data, or clinical interventions. It uses publicly available image datasets and offline computational experiments.

Competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. WYSAWIS, cited in the related work, shares authors with this study and is identified explicitly. It is used as qualitative published context, not as an experimental baseline or as evidence for the numerical claims reported here.

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Data availability. BOSSBase 1.01 is publicly available from its original distributor at https://agents.fel.cvut.cz/boss/; Caltech-101 is available from CaltechDATA at https://data.caltech.edu/records/mzrjq-6wc02; and CIFAR-100 is available at https://www.cs.toronto.edu/~kriz/cifar.html. The datasets are not redistributed in the project repository. Deterministic partition seeds and consolidated CSV/JSON results accompany the reproducibility package.

Code availability. The NARCIS implementation, attack definitions, evaluation scripts, tests, consolidated result tables, plotting code, manuscript sources, and exact random seeds are publicly available at https://github.com/EkodeckStephane/NARCIS. Trained checkpoints are not redistributed; the repository provides the commands required to retrain the models.

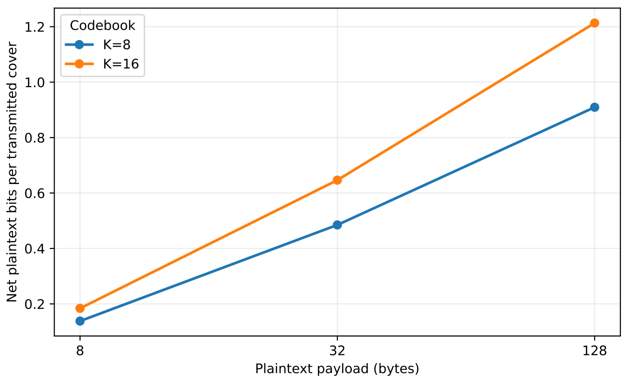

Figure 3: Net plaintext rate for the implemented protected frame. The three-bit and four-bit operating points correspond to $K = 8$ and $K = 16$.

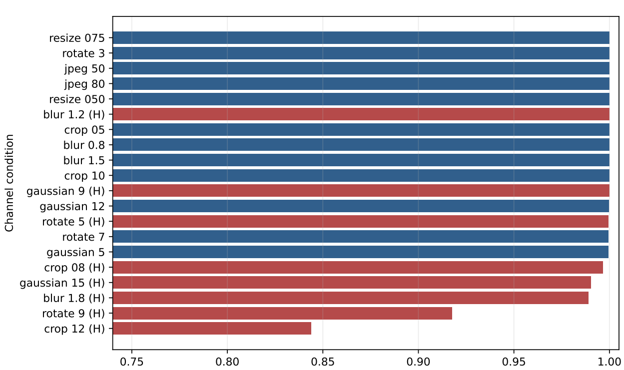

Figure 4: Mean BOSSBase symbol accuracy over five seeds and ten messages per seed. Holdout strengths were excluded from codebook qualification. Complete-message success was 100% for every condition.

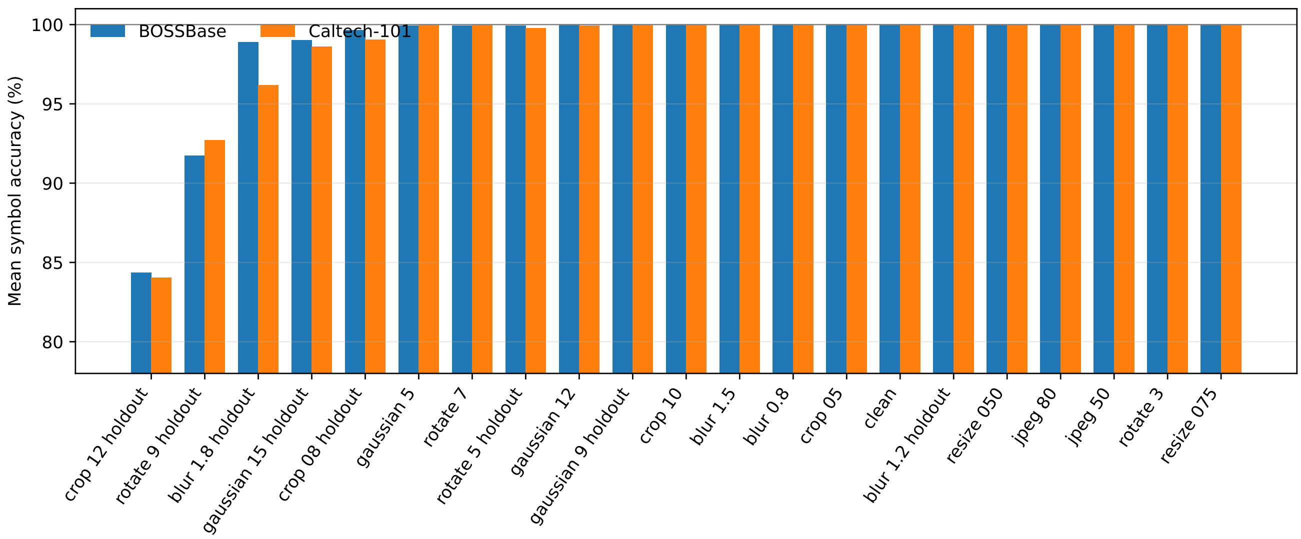

Figure 5: Mean symbol accuracy across the five BOSSBase and five Caltech-101 partitions. Calibration and holdout conditions use the same declared strengths; all 1,575 complete-message trials succeeded.

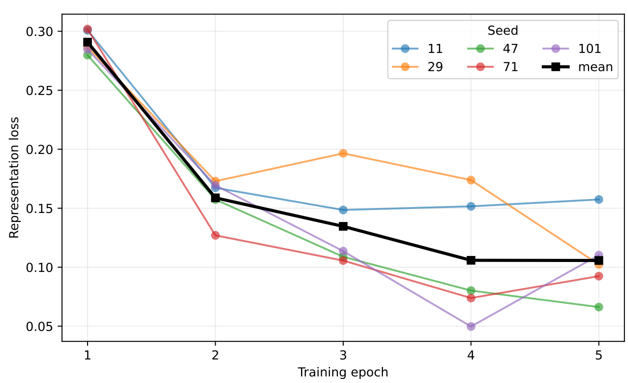

Figure 6: BOSSBase training loss for the five deterministic partitions.

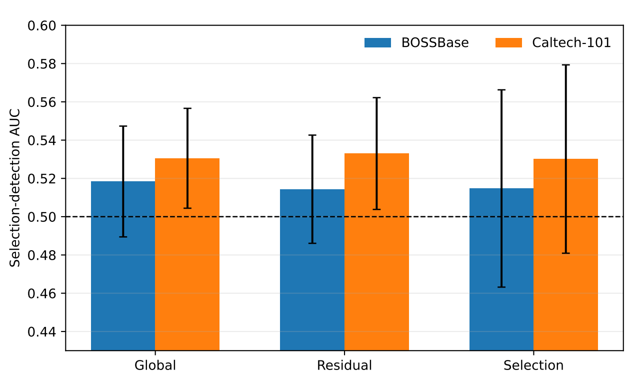

Figure 7: Mean selection-detection AUC and 95% Student-t intervals over five partitions per dataset. The dashed line denotes chance performance.

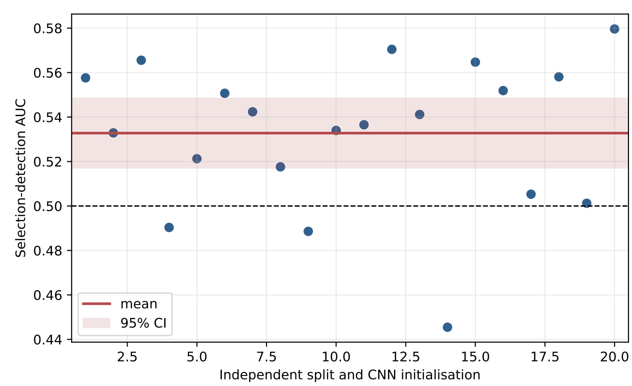

Figure 8: Selection-CNN AUC over 20 independent balanced subsamples, splits, and initialisations on Caltech-101 seed 47. The shaded band is the 95% Student-t interval for the repeated fits.

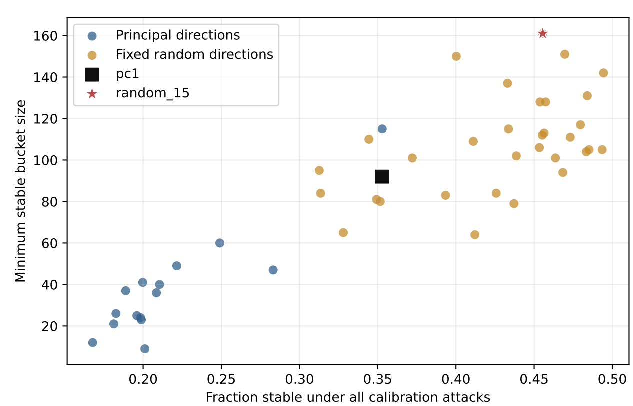

Figure 9: Calibration-only projection candidates on the full BOSSBase seed-11 index. The selected direction maximises minimum stable-bucket occupancy under the declared lexicographic rule; holdout attacks are not used for selection.

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